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GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	**Specification for Pipe Supports**
Document Number	EA-BFPCL-GPMC-GGPL-FEED- MPP-SPC-011
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Specification for Pipe Supports

R01	24.05.22	Issued for Review	E. Uwaoma	E.Nwachukwu	K. Yusuff	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R01	24.05.22	First Issue for Clients Review
R02		
A01		

Review Code & Status (to be filled by Client)		
☐ **Code 1**	**Approved –** Further work may proceed.	
☐ **Code 2**	**Approved with comments.** – Incorporate Comments. Up-Rev & Resubmit. Work May Proceed	
☐ **Code 3**	**Reviewed** – Incorporate Comments. Up-Rev & Resubmit. Work May Not Proceed	
☐ **Code 4**	**Retained** for information only	
☐ **Code 5**	**Review Not Required** – Do not Resubmit	
Date:	**Position:**	**Sign:**

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited		
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A	
Facility:	Gas Gathering Pipelines & Associ-ated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED- MPP-SPC-011 MPP-SPC-011	

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Classes
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-014	General Piping Arrangement Drawing - BBOU
EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-015	General Piping Arrangement Drawing - ELEP
EA-BFPCL-GPMC-GGPL-FEED-MPP-DWG-016	General Piping Arrangement Drawing - AKAB
EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-002	Piping Stress Analysis and Report - BBOU
EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-003	Piping Stress Analysis and Report - ELEP
EA-BFPCL-GPMC-GGPL-FEED-MPP-RPT-004	Piping Stress Analysis and Report - AKAB

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1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

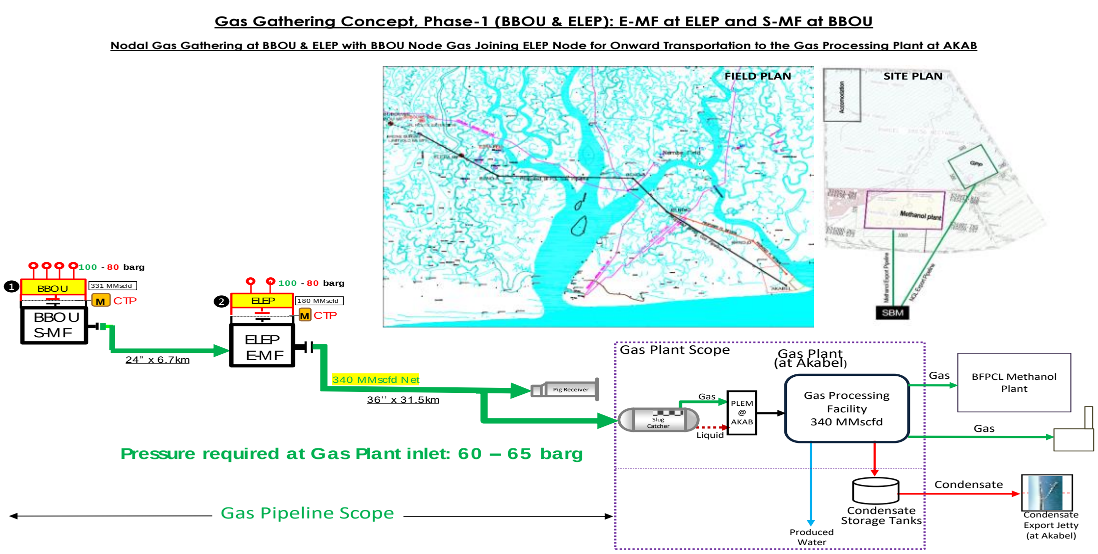


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This document presents the technical requirement for the design, selection and location of pipe support for the in-plant piping general arrangement associated with BFPCL Phase 1 pipeline project.

This specification covers the selection of pipe support shoes and features for all associated piping that shall be installed aboveground. The design of support structures is not covered by this specification.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP31.38.01.29-Gen.	Pipe Support
DEP31.38.01.24-Gen.	Piping – Engineering And Layout Requirements
DEP 31.38.01.31-Gen	Shop And Field Fabrication Of Piping
DEP02.00.00.10-Gen.	Preparation Of Technical Drawings
DEP01.00.09.10-Gen.	Operational tagging requirements
DEP 30.48.00.31-Gen	Protective Coatings for Onshore and Offshore Facilities
DEP 31.38.01.26-Gen	Piping-Pipe Stress Analysis Requirements
S34.034	Standard lined pipe shoe type S7 thru S12
S34.035	Standard welded pipe shoe type S29 thru S34
S44.033	Standard pipe shoe type S1 thru S6
S44.032	Typical bracings for small bore branches of piping (e.g. pressure instrument connection)
S44.030	Typical bracings for small bore branches of piping (e.g. orifice instrument connection)
S44.041	Clamped base support type CB1 thru CB9
S44.036	Standard guide type G1 thru G10
S44.039	Standard Base Support

ASME B1.1	Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B1.20.1	Pipe Threads, General Purpose
ASME B18.2.1	Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	Square and hexagonal nuts
ASME B31.3	Process Piping
BS 3974	Specification for pipe supports. Pipe hangers, slider and roller type supports
BS 4360	Specification for weldable structural steels
MSS SP-58	Piping hangers and supports-Material design and manufacture
MSS SP-69	Pipe Hangers and Support - Selection and Application
MSS SP-89	Piping hangers and supports-Fabrication and installation practice

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Environmental Conditions

3.1. Environmental Data

Refer to EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002 Basis of Design.

Welding processes of the pipe supports shall put into consideration the environmental conditions in which the welding activities shall take place and where the system is meant to operate.

4. General Requirement for Piping support

4.1. General

The selection of pipe support shall be based on all concurrently acting loads transmitted into such supports under worst case load combination scenario. Susceptible loads on the associated piping in this project include weight effects, loads introduced by service pressures and temperatures, thrust force from pressure relief devices, force from possible slug flow, wind, and displacement strain. These load types are defined in para. 301 of ASME B31.3

Technical requirements for design of supporting structural or concrete elements is specified in specification for structural steel works

Supporting elements shall be directed toward preventing the following:

- Piping stresses in excess of those permitted in the Code
- Leakage at joints
- Excessive thrusts and moments from relief valve
- Excessive stresses in the supporting (or restraining) elements
- Resonance with imposed or fluid-induced vibrations
- Excessive interference with thermal expansion and contraction in piping which is otherwise adequately flexible
- Unintentional disengagement of piping from its supports
- Excessive piping sag, normally more than ½ the diameter of pipe, process piping

Supports shall be standardized as far as practical and shall be selected from the listed drawings in DEP 31.38.01.29-Gen

Supports shall bear an identification number in accordance with the tagging philosophy of DEP 01.00.09.10-Gen.

Pipe support identification numbers shall be shown clearly on piping general arrangement drawings and on the Piping Isometric drawings.

Supports and supporting structures shall be able to sustain the hydrostatic test load. If this is not economical, temporary supports may be used to support the pipes during hydrotest and replaced with the permanent ones after the operation

Weld-on attachments of pipe supports should be welded to the pipe during prefabrication of the pipe spools.

Supporting one piping system hanging from another piping system shall be avoided.

Pipe supports shall not block access for operation and maintenance.

Allowable stresses for materials commonly used in the design of pipe supports are listed in Tables 2 and A1 of MSS SP 58. Allowable stresses for materials not listed in Tables 2 and

A1 of MSS SP 58, with known physical properties shall be determined as the lower of the following values:

- 1/4 of minimum tensile strength at service temperature
- 5/8 of minimum yield strength at service temperature.

5. Support Selection

5.1. General

Standard pipe support type and location shall be selected based the rules set forth in this document. Stress Analysis shall be carried out on the supported piping systems to determine the resultant concentrated and distributed load, stresses and moments in other to refine the support types and locations.

Below are characteristics associated with the Piping system to be considered in

A design temp of 60ºC and below with few lines operating at sub-zero temperature such as overpressure relief line and depressurization lines.

- Thermal expansion/contraction from temperature variations.
- Piping materials to be compatibility with materials of piping attachments and supporting structures.
- Equipment Handling and safety issues.
- Non insulated lines
- Drain lines and flare header that require sloping.
- Thrust forces from pressure relief valves
- Possible slug flow in the multiphase pipeline system.
- Concentrated load at locations where valves and other heavy items are installed.

5.2. Piping systems

The following are the piping systems (with different arrangements that shall encountered) in the project where supports shall be installed.

5.2.1. Non-insulated pipes

All piping system in the project are Non-insulated pipes. Therefore, the pipes shall be having direct contact with the support beams with or without neoprene pads.

Pipe shoes shall be used for the following:

- Carbon steel pipes with a wall thickness less than schedule 20.
- The pipe requires a slope. Where pipe shoes are used to support sloping pipe, the height of pipe shoes measured from the underside of the pipe shall be a maximum 400 mm (16 in).

For piping systems where only small forces on supports can be expected (e.g., instrument air systems), U-bolts may be used. Use of U-bolt as fixed points shall be avoided because their grip is not reliable and damages the coating system.

5.2.2. Small bore piping

Where there is need to reinforce the mechanical strength of small-bore piping exposed to flow induced vibration and self-weight, supports design shall be as shown on Standard Drawings S 44.030, S 44.031 and S 44.032 shall be considered.

5.2.3. Vertical pipes

Vertical pipes shall be supported using a clamped supports applied on vertical section at the upper half of the center of gravity. It shall be executed with weld-on shear lugs, to avoid the pipe slipping under the clamp. Vertical pipes may also be supported at the base using pipe dummy legs (Trunnion)

5.2.4. Supports for tubing

Tubing for instrumentation shall be supported along steel beams or trays according to the support spans recommended by Manufacturers/Suppliers of the tubing system.

5.2.5. Support for Valves

Valves sizes less the 12 inches shall be supported on the adjoining piping as much as possible. Where it is absolutely necessary, to support the valve directly (example where the valve is at the end of the piping system with a blind flange) bracing shoes bolted at the flanges shall be used to support the valves on the concrete foundation.

Valves 12 inches and above shall have a concrete base support constructed directly under the valve.

Valve supports legs or saddles shall be supplied and installed on valves of 12 inches and above.

The supports must allow access to body drain plugs where fitted.

Refer to EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-003- Specification for Piping valves.

5.3. Types of Pipe Supports

The following standard support types shall be used for this project.

5.3.1. Clamped Type Supports

Clamped supports require more maintenance than welded supports. To prevent galvanic corrosion, carbon steel clamps on pipes of other Carbon Steel materials shall be separated from the pipe by using a layer of synthetic insulating material (Neoprene) between the clamp and the pipe.

Clamped supports on hot pipes or on pipes with variable temperature may loosen because of the difference in temperature between the pipe and the clamp. In those situations, weld-on shear lugs should be applied in order to avoid slipping of the pipe under the clamp. The paint of the pipe shall be repaired after welding of shear lugs.

The position of clamped supports may be adjusted during construction to correct mismatches between the location of the support and the location of the supporting structure. These field modifications shall only be minor corrections and shall not lead to an increase of stresses and/or moments in the piping system and/or in the connecting nozzles.

5.3.2. Pipe shoes welded on pipe

Welded type supports are less susceptible to corrosion than clamped type supports and shall be used as pipe supports in order to reduce corrosion of contact area between pipe and the support. Provides a fixation where all six degrees of freedom are restraint if it is welded at the base.

Welded type pipe supports shall not be used on Piping requiring post weld heat treatment. clamped type pipe supports shall be used.

5.3.3. Restraints

Any device that prevents, resists, or limits the free thermal movement of piping. Restraints shall be used in combination with pipe shoes and clamps to provide a required support.

5.3.4. U-Bolt

U-Bolt shall be used to provide restraint for small bore piping typically DN50 (NPS 2) and below.

5.3.5. Temporary Support

Temporary supports could be used during transportation and hydro test activities and shall be removed after the temporary activity

6. Pipe Support Numbering

Each pipe support shall be tagged with a unique identification number in accordance with project procedure.

The general format for tag number will be as shown below:

AA – XXYY - ZZZ

Where AA = Support Type

XX = Sector Code

YY = Deck

ZZZ = Three digits sequential number

6.1. Standard Pipe Support Tag Numbers

PS - XXYY - ZZZ

Pipe Support

Sector code/Deck

7. Support Location

Pipe supports should be positioned on the piping system as follows:

- On pipe rather than valves and flanges for valves less than 12”
- All piping 2" NPS and above.
- All stress critical lines (irrespective of line size) as indicated in the critical line list.
- As close as possible to heavy components such as relief valves and isolation valves in particular for high pressure rating and actuated valves for all pressure ratings.
- At the positions where the vertical movements are minimal
- Supports shall be located as close as practicable to changes in direction, but allow adequate flexibility for thermal expansion and settlement of the pipe
- The location and type of pipe supports shall be in accordance with the stress analysis outcome for identified critical lines.
- Good engineering judgement and rule of thumb shall also be applied in positioning supports for both critical and non-critical lines.
- Where vertical movements are anticipated at pipe supports due to thermal expansion and settlement of piping, resting type supports with guides or stopper shall also be used.

7.1. Layout considerations for pipe support.

Pipes shall be grouped together to take advantage of singular pipe support arrangement. To maintain common support levels, the back or underside of pipes (or underside of supports) shall be on the same plane or level, irrespective of pipe size.

One anchor support shall be provided for pipe length spanning for more than 100m.

When slope ratio is specified for a line, pipe shoe support of varying height shall be used to achieve sloping and shall be designed to withstand buckling loads. The height of pipe shoes measured from the underside of the pipe shall be a maximum 400 mm (16 in).

Control valve manifold where high pressure drop is expected e.g. the depressurization line shall be supported rigidly as to sustain hydraulic forces due to differential pressure across control valve and to prevent the flow-induced vibration of piping system.

Piping systems in pipe racks or pipe tracks shall be guided at intervals of 2x the pipe support span to guard against the lateral movements due to thermal expansion, including solar radiation.

Sufficient space shall be provided to facilitate support assembly installation, inspection and maintenance

7.2. Support Spacing

Allowable spacing between two support on a straight line otherwise called spans are based on a maximum bending stress of 41.37 Mpa (6000 PSI) or a maximum deflection of 12.7 MM (0.5in) or 1/2 of pipe diameter using simple beam analysis method.

Pre-determined Span values tabulated in Appendix A1 of DEP 31.38.01.29-Gen shall be applicable in this project. Correction shall be made to this span based on the result of the stress analysis. For straight run pipe length only. When change of direction in a horizontal plane occur, the straight run length is multiplied by a shape factor. See fig 1.

8. Welding of Pipe Supports.

Field welding to pipes for pipe supporting purposes should be limited as far as practical.

Field welding for pipe support purposes shall not be performed on pipe materials requiring post weld heat treatment. Welding shall be performed in the workshop before post weld heat treatment. All welds of support elements and of supports to piping shall be continuous. Welding of supports shall be completed prior to hydrostatic testing, flushing, chemical cleaning or blowing.

All welded type pipe supports shall have pipe support component materials of the same grade as the main pipe

Components of clamped type pipe supports or U-bolts that are in direct contact with the pipe are the same basic grade as the base pipe material

Polymeric materials used in the construction of pipe supports are compatible with the maximum/minimum exposure temperatures.

9. Material Selection

All welded type pipe supports shall have pipe support component materials of the same grade as the main pipe

Components of clamped type pipe supports or U-bolts that are in direct contact with the pipe are the same basic grade as the base pipe material

Polymeric materials used in the construction of pipe supports are compatible with the maximum/minimum exposure temperatures.

10. Installation

Pipe supports shall be installed in accordance with the piping layout drawings and pipe support detail drawings.

Piping shall be anchored and guided before pressure testing.

Base supports shall have welded end plates.

For sliding supports, bearing surfaces shall be sufficiently clean to ensure unrestricted movement.

Safe loads on anchor bolts used for connecting supports to concrete shall be based on the loading capacity of the concrete.

11. Marking and Painting

The metallic surfaces of supports shall be protected in according with the project Painting and Coating Specification.

Each standard support component shall be die-stamped with its size and identification mark.

12. Recommended Pipe span

For pipe resting on more than two supports (Ref: DEP 31.38.01.26-Gen) the standard span is as shown in Figure 1 below

Figure 1: Pipe Support Span.

A.1 CARBON STEEL AND HEAVY WALL STAINLESS STEEL

1. For systems where (A.1, Item 2) does not apply, refer to Table A.1 for span data applicable to the following:
   a. carbon steel pipes for sizes DN 50 (NPS 2) and above, STD wall and heavier, with a maximum temperature of 350 °C (660 °F);
   b. austenitic stainless steel pipes, schedule 40S and heavier, DN < 400 (NPS < 16), with a maximum temperature of 350 °C (660 °F);
   c. carbon steel pipes for sizes DN 15 (NPS ½) to DN 40 (NPS 1 ½), schedule 80 wall and heavier, with a maximum temperature of 350 °C (660 °F).
2. For free draining pipes with a slope shallower than 1:240, Table A.1 shall not be used.
3. Spans shown in Table A.1 are based on straight pipe; other configurations shall be multiplied by a shape factor in accordance with the sketch shown in Table A.1.

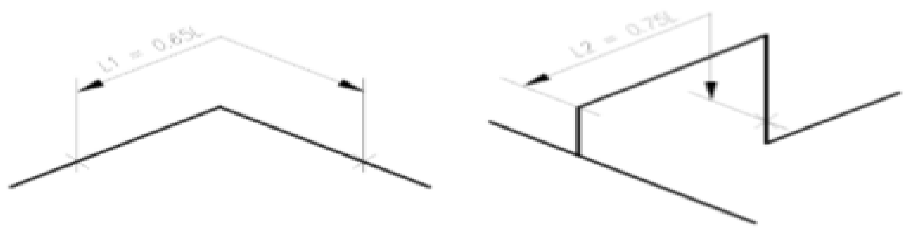
Table A.1 Maximum spans for carbon steel and heavy wall stainless steel

Pipe size	Maximum span[1]			
	Vapour service		Liquid service	
	Bare	Insulated[2]	Bare	Insulated[2]
DN 15 (NPS ½)	900 mm (3 ft)	800 mm (2 ½ ft)	900 mm (3 ft)	800 mm (2 ½ ft)
DN 20 (NPS ¾)	1400 mm (4 ½ ft)	1200 mm (3.9 ft)	1400 mm (4 ½ ft)	1200 mm (3.9 ft)
DN 25 (NPS 1)	3600 mm (11.8 ft)	2300 mm (7.5 ft)	3450 mm (11.3 ft)	2250 mm (7.3 ft)
DN 40 (NPS 1 ½)	3600 mm (11.8 ft)	3000 mm (9.8 ft)	3450 mm (11.3 ft)	2800 mm (9.1 ft)
DN 50 (NPS 2)	3600 mm (11.8 ft)	3450 mm (11.3 ft)	3450 mm (11.3 ft)	3300 mm (10.8 ft)
DN 80 (NPS 3)	6550 mm (21.4 ft)	4600 mm (15 ft)	5450 mm (17.8 ft)	4200 mm (13.7 ft)
DN 100 (NPS 4)	7500 mm (24.6 ft)	5550 mm (18.2 ft)	6100 mm (20 ft)	4900 mm (16 ft)
DN 150 (NPS 6)	9150 mm (30 ft)	6800 mm (22.3 ft)	7100 mm (23.2 ft)	5800 mm (19 ft)
DN 200 (NPS 8)	10500 mm (34.4 ft)	8050 mm (26.4 ft)	7950 mm (26 ft)	6700 mm (21.9 ft)
DN 250 (NPS 10)	11800 mm (38.7 ft)	9050 mm (29.6 ft)	8700 mm (28.5 ft)	7400 mm (24.2 ft)
DN 300 (NPS 12)	12900 mm (42.3 ft)	9800 mm (32.1 ft)	9150 mm (30 ft)	7800 mm (25.5 ft)
DN 350 (NPS 14)	15150 mm (49.7 ft)[3]	11850 mm (38.8 ft)	10850 mm (35.5 ft)	9300 mm (30.5 ft)

Pipe size	Maximum span[1]			
	Vapour service		Liquid service	
	Bare	Insulated[2]	Bare	Insulated[2]
DN 400 (NPS 16)	16250 mm (53.3 ft)[3]	12850 mm (42.1 ft)	11200 mm (36.7 ft)	9750 mm (31.9 ft)
DN 450 (NPS 18)	17250 mm (56.5 ft)[3]	13750 mm (45.1 ft)	11500 mm (37.7 ft)	10150 mm (33.3 ft)
DN 500 (NPS 20)	18200 mm (59.7 ft)[3]	14450 mm (47.4 ft)	11750 mm (38.5 ft)	10400 mm (34.1 ft)
DN 600 (NPS 24)	18950 mm (62.1 ft)[3]	16050 mm (52.6 ft)	12150 mm (39.8 ft)	10950 mm (35.9 ft)
DN 750 (NPS 30)	21000 mm (68.9 ft)[3]	17500 mm (57.4 ft)	13100 mm (43 ft)	11500 mm (37.7 ft)
DN 900 (NPS 36)	22700 mm (74.5 ft)[3]	18500 mm (60.7 ft)	13700 mm (45 ft)	12500 mm (41 ft)
DN 1050 (NPS 42)	23400 mm (76.8 ft)[3]	19500 mm (64 ft)	14300 mm (47 ft)	13000 mm (42.6 ft)
DN 1200 (NPS 48)	25000 mm (82 ft)[3]	20500 mm (67.2 ft)	14600 mm (48 ft)	13400 mm (44 ft)

NOTES:

1. Spans are based on straight pipe. A shape factor, as illustrated in the following sketches, applies for other configurations



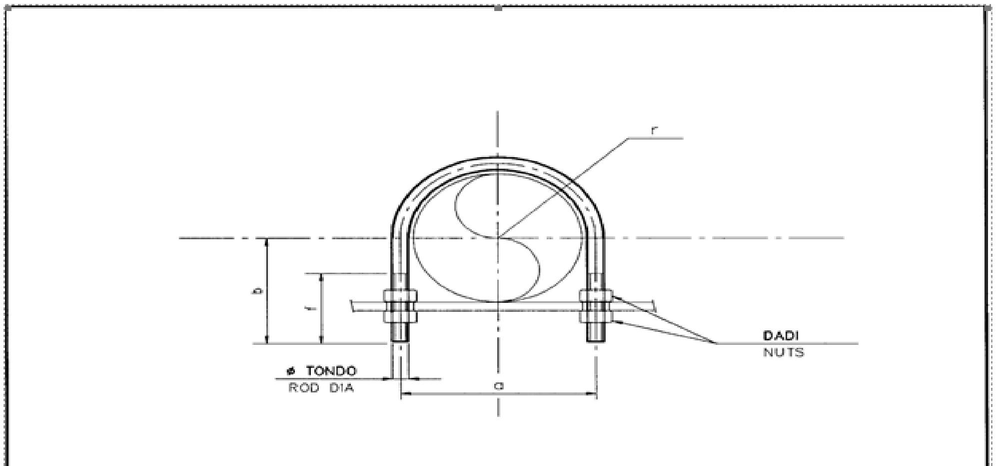
2. The weight of insulation and sheeting is based on insulation thickness varying from 50 mm (2 in) for DN 15 (NPS ½) to 160 mm (6 ¼ in) for DN 1200 (NPS 48) and a density of 190 kg/m³ (12 lb/ft³).
3. Spans limited by deflection. All other spans are limited by longitudinal bending stress.

13. Support Standaeds.

Collection of support types that are applicable to this Project are presented below.

The detailed design drawing and MTO of the supports shall be compiled in detailed engineering.

Fig 2: "U"-Bolts Guiding for uninsulated pipes.



GRANDEZZA SIZE DN/ND	ø TONDO ROD DIA	r	a	b	f	SVILUPPO DEVELOPMENT			PESO WEIGHT Kg
1/2"	M6	12	30	40	37	130			0.1
3/4"	M6	15	36	45	40	150			0.1
1"	M6	18	42	50	40	170			0.1
1 1/2"	M8	25	58	60	45	220			0.1
2"	M8	31	70	65	45	240			0.1

U-bolt supports may be used in exceptional situations and shall be restricted to pipes of NPS ≤ 2 inch.

Fig 3. Standard welded pipe shoe (Ref: S34.035)

Full Welding of shoe attachment shall not penetrate pipe

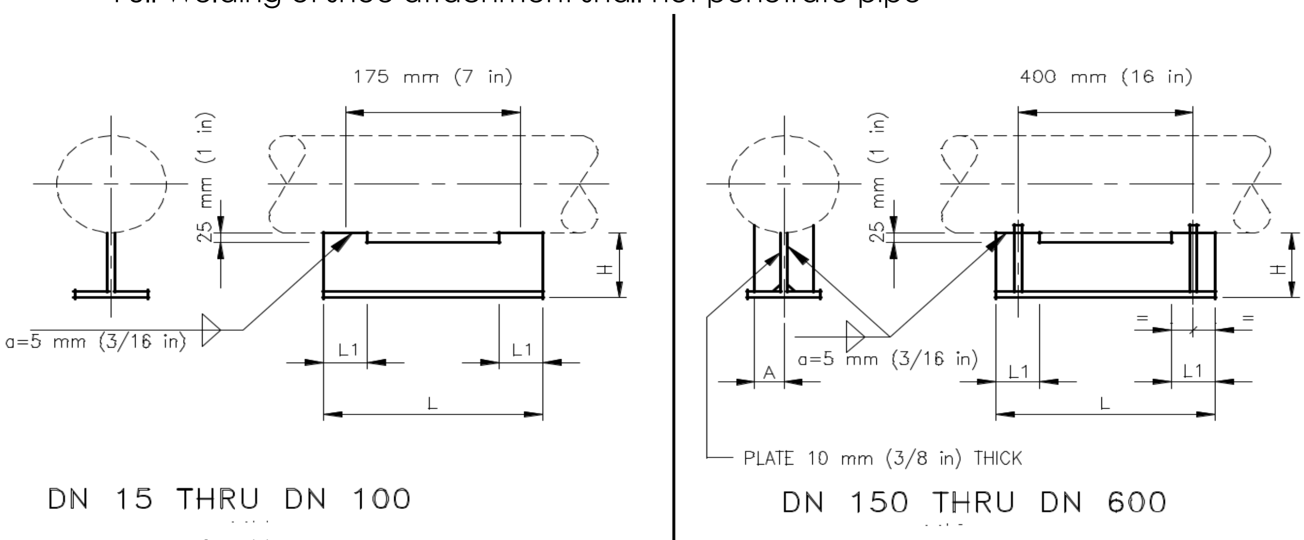


Fig 4: Standard Pipe Shoe welded on clamp (Ref. S44.033)

Structural members to be trimmed as required.

All clamps shall be arranged with the clamping bolts at an orientation as indicated below

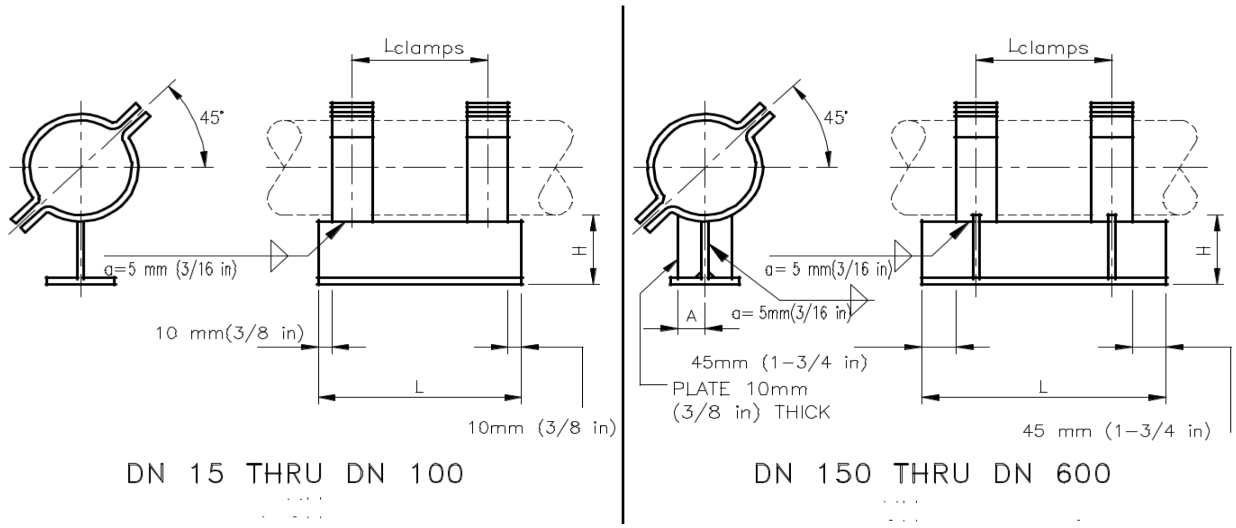


Fig 5: Typical brazing branches of piping (Ref S44.032)

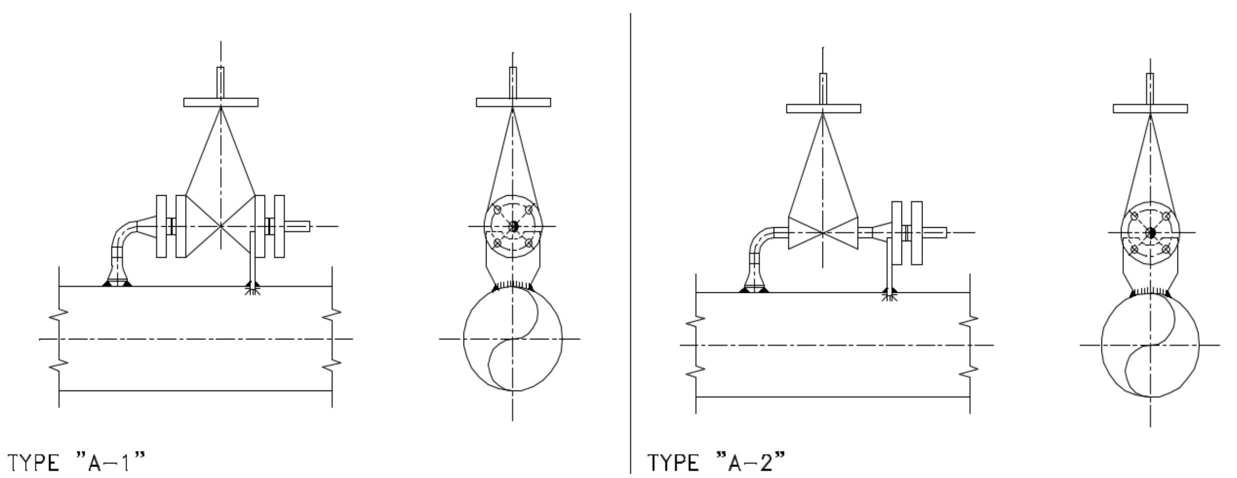


Fig 6: Typical brazing branches of piping (Ref S44.030)

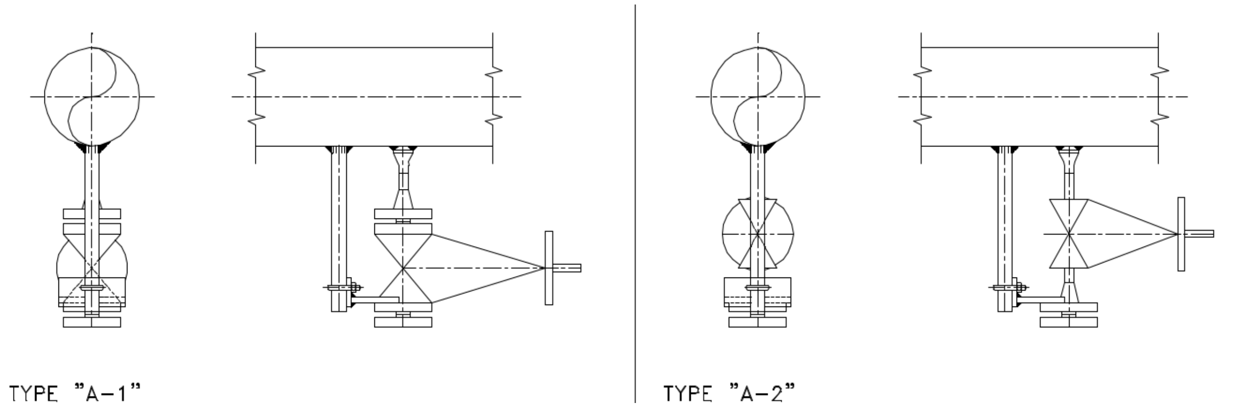


Fig 7: Clamped Dummy/Baseplate Support (Ref: S44.041)

Wall thickness of dummy in accordance with relevant piping class

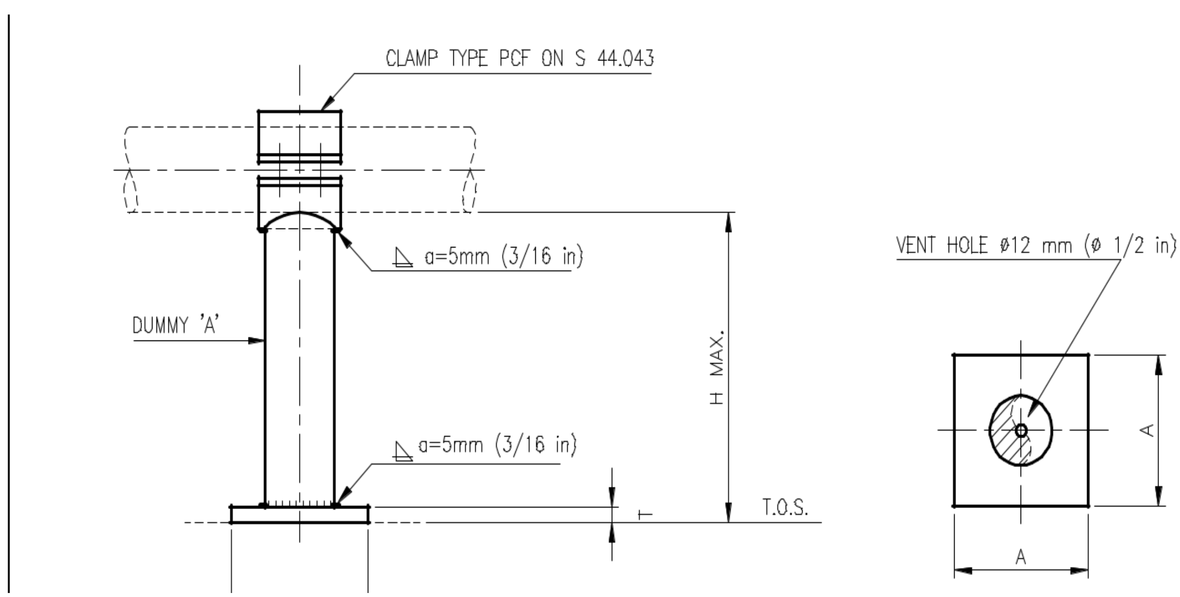


Fig 8; Example of Heavy-duty pipe saddle that shall be used for Pig trap (dimension here are for indication only)

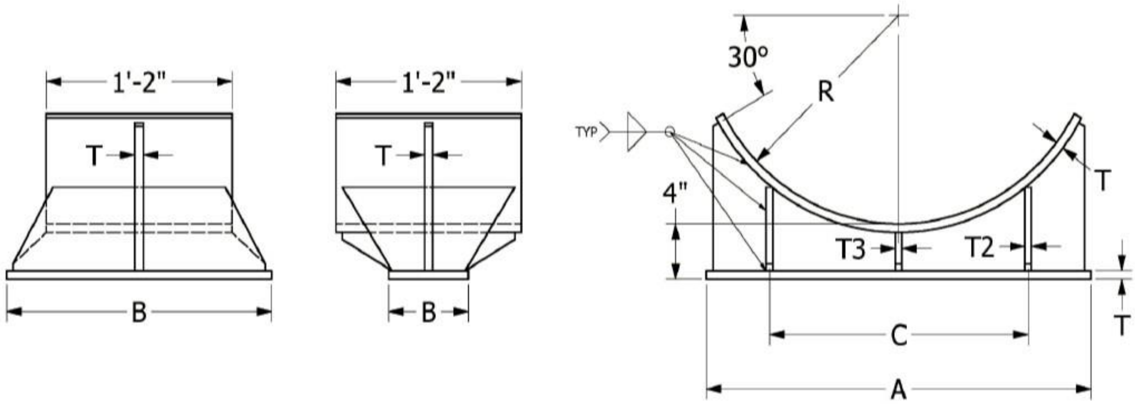
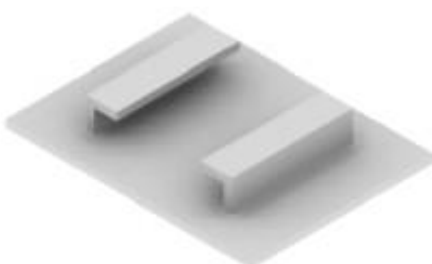
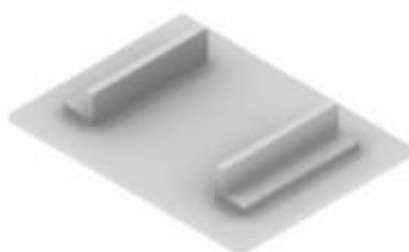


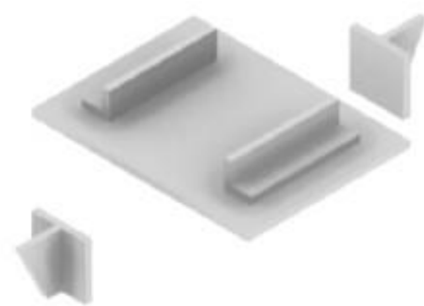
Fig 9; Restraint options (real image)



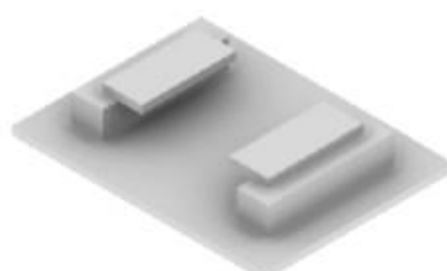
1Angle guide (Type 02)



2Angle guide (type 02)



3Angle guide and line stop



4Angle guide and lift stop

Fig 10; Supports for vertical pipes

a) Pipe clamp for vertical pipes

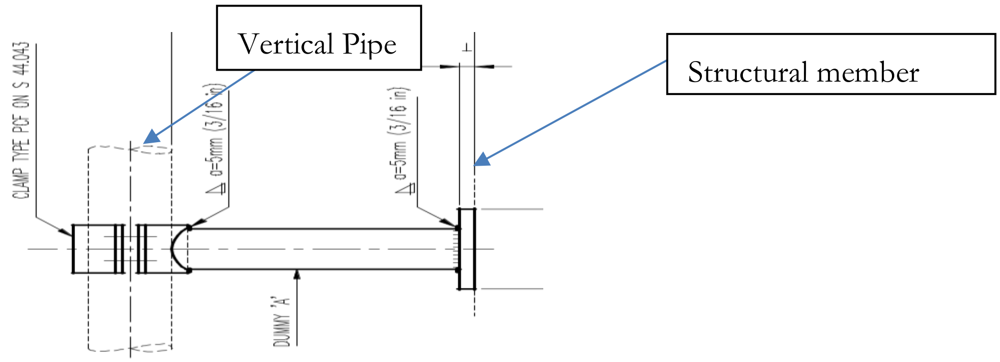


Fig 11; Standard Base Support; Dummy leg (Ref S44.039)

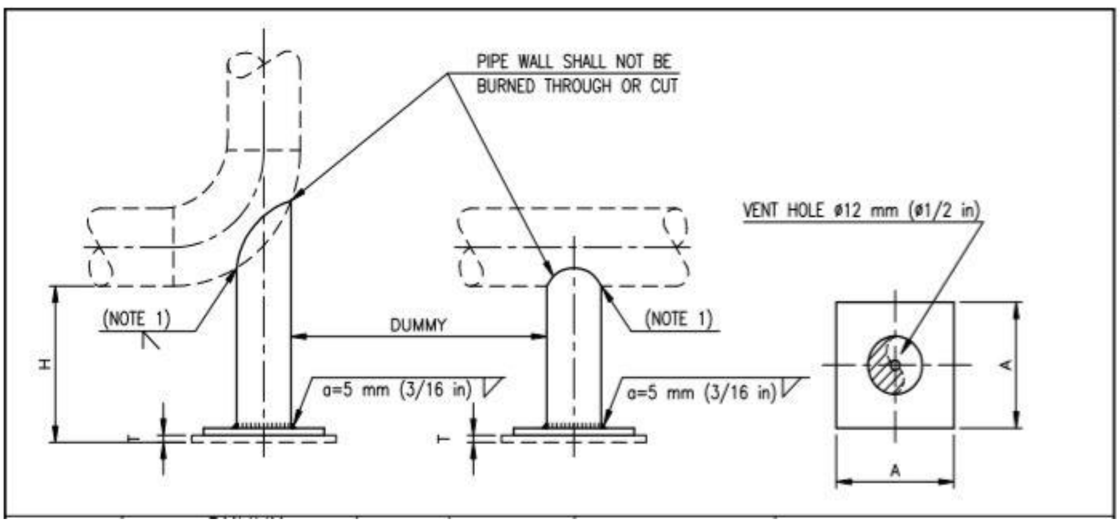
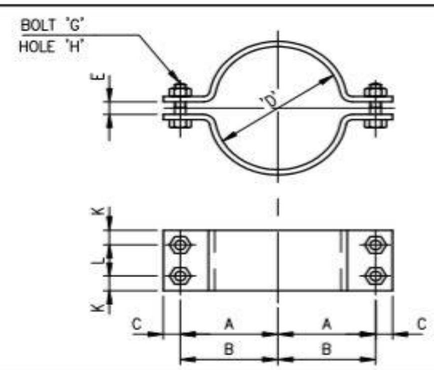


Fig 12; Standard Clamp Heavy Type NPS ½ through NPS 24 (Ref S 44.043)



TYPE	DN xx NPS (xx)	A mm (in)	B mm (in)	C mm (in)	D mm (in)	E mm (in)
PCF	DN 40 (NPS 1-1/2)	42 (1-5/8)	55 (2-3/16)	13 (1/2)	49 (2)	12 (1/2)
	DN 50 (NPS 2)	56 (2-3/16)	72 (2-7/8)	16 (5/8)	61 (2-3/8)	
	DN 80 (NPS 3)	71 (2-3/4)	87 (3-7/16)		90 (3-1/2)	16 (5/8)
	DN 100 (NPS 4)	90 (3-1/2)	110 (4-3/8)	20 (3/4)	115 (4-1/2)	20 (3/4)
	DN 150 (NPS 6)	124 (4-7/8)	146 (5-3/4)	22 (7/8)	169 (6-5/8)	22 (7/8)
	DN 200 (NPS 8)	152 (6)	174 (6-7/8)		220 (8-11/16)	26 (1)
	DN 250 (NPS 10)	189 (7-7/16)	218 (8-5/8)	29 (1-3/16)	274 (10-3/4)	
	DN 300 (NPS 12)	214 (8-7/16)	243 (9-5/8)		325 (12-13/16)	
	DN 350 (NPS 14)	235 (9-1/4)	270 (10-5/8)	35 (1-3/8)	357 (14)	28 (1-1/8)
	DN 400 (NPS 16)	260 (10-1/4)	295 (11-5/8)		406 (16)	
	DN 450 (NPS 18)	300 (11-7/8)	335 (13-3/16)		457 (18)	32 (1-1/4)
	DN 500 (NPS 20)	324 (12-3/4)	359 (14-1/8)		508 (20)	36 (1-7/16)
	DN 600 (NPS 24)	381 (15)	422 (16-5/8)	41 (1-5/8)	610 (24)	41 (1-5/8)

TYPE	DN xx NPS (xx)	K mm (in)	L mm (in)	HOLE, H mm (in)	BOLT, G mm (in)	STEEL SIZE mm (in)
PCF	DN 40 (NPS 1-1/2)	35 (1-3/8)		10 (7/16)	M8 X 35 (UNC 5/16 X 3/8)	70 X 5 (2-3/4 X 3/16)
	DN 50 (NPS 2)			14 (9/16)	M12 X 50 (UNC 1/2 X 2)	70 X 8 (2-3/4 X 5/16)
	DN 80 (NPS 3)	40 (1-9/16)				80 X 8 (3-3/16 X 5/16)
	DN 100 (NPS 4)	55 (2-3/16)			M12 X 60 (UNC 1/2 X 2-3/8)	110 X 10 (4-3/8 X 7/16)
	DN 150 (NPS 6)	25 (1)	90 (3-1/2)		M12 X 70 (UNC 1/2 X 2-3/4)	140 X 10 (5-1/2 X 7/16)
	DN 200 (NPS 8)		150 (6)			200 X 10 (8 X 7/16)
	DN 250 (NPS 10)	40 (1-9/16)	170 (6-3/4)	18 (3/4)	M16 X 80 (UNC 5/8 X 3-3/16)	250 X 10 X (10 X 7/16)
	DN 300 (NPS 12)		220 (8-11/16)			300 X 16 (12 X 5/8)
	DN 350 (NPS 14)	50 (2)	250 (9-7/8)	22 (7/8)	M20 X 85 (UNC 3/4 X 3-3/8)	350 X 16 (14 X 5/8)
	DN 400 (NPS 16)		300 (11-7/8)			400 X 16 (16 X 5/8)
	DN 450 (NPS 18)		350 (13-3/4)		M20 X 100 (UNC 3/4 X 4)	450 X 16 (18 X 5/8)
	DN 500 (NPS 20)				M20 X 120 (UNC 3/4 X 4-3/4)	
	DN 600 (NPS 24)					